

Leonardo Herrera

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Spring, TX

Portfolio: herrera2.github.io/Personal-Website/index.html

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Work Experience

Manufacturing Design Engineer

May 2025 – Present

Duke Manufacturing – CounterCraft

Houston, TX

- Design and release sheet-metal structural components in SolidWorks, ensuring dimensional integrity and manufacturability from concept through fabrication in a production environment.
- Develop sheet-metal flat patterns, bend allowances, and CNC nesting layouts to optimize material usage, reduce scrap, and improve production efficiency.
- Implemented a parametric universal mullion design framework in SolidWorks to standardize internal support structures across custom countertop assemblies, reducing modeling time by ~15-20%.
- Apply tolerance stack-up analysis to control variation and prevent downstream assembly misalignment and rework.
- Partner directly with fabrication and shop-floor teams to troubleshoot structural nonconformances, perform root cause investigations, and implement corrective design or process improvements.
- Interpret engineering drawings in AutoCAD to ensure compliance with manufacturing and quality standards.

Mechanical Design & Test Engineering Intern

March 2022 – September 2023

NASA Johnson Space Center – CACI (Contractor), PVAMU (Subcontractor)

Houston, TX

- Supported spacecraft docking certification campaigns for ISS and Lunar Gateway programs under the Six Degree of Freedom Dynamic Test System (SDTS), using a 6DOF robot to validate interface performance and load response.
- Contributed to full test lifecycle operations including test article integration, instrumentation setup (force/moment sensors, optical tracking, laser metrology), configuration control, and real-time test execution in a test environment.
- Served as test operator during dynamic docking simulations (SpaceX Crew Dragon and JAXA HTV-X), executing closed-loop force and position control runs in coordination with NASA engineers and contractors.
- Designed and upgraded mechanical hardware supporting test infrastructure, including sensor mounts, structural interfaces, and electronics fixtures; performed stress and modal analysis for stability and vibration compliance.
- Developed mechanical and control models of the 6DOF test system using MATLAB and Simulink, performing system identification, transfer function derivation, and chirp analysis to support actuator and torque characterization.
- Supported the Systems Engineering Simulator (SES) program by developing 3D CAD models and detailed engineering drawings for hardware installations, performing interference checks, and coordinating with systems engineers.

Education

Master of Science in Engineering

Graduated December 2023

Prairie View A&M University, Prairie View, TX

GPA: 3.80

- Thesis: Control Filter Assessment on Six Degree of Freedom Stewart Platform (Publication Pending)

Bachelor of Science in Mechanical Engineering

Graduated May 2022

Prairie View A&M University, Prairie View, TX

GPA: 3.67

- Senior Project: Designed, modeled, and built a prototype of a Lightweight Deployable Shield for Ball Aerospace.

Awards

Elite Team Award, NASA

August 2, 2023

Elite Team Award, NASA

September 14, 2022

Skills

CAD & Design Analysis: SolidWorks | Creo Parametric | Siemens NX | AutoCAD | FEA (Stress, Modal) | Creo Simulate | GD&T | Engineering Drawings (ASME Y14.5) | Ansys | MATLAB | Simulink

Test & Integration: Robotic System Operation | Test Equipment Setup | Mechanical Fixture Design | Hardware Validation | Instrumentation Setup (Force/Torque Sensors, Optical Tracking, Laser Metrology)

Manufacturing & Production: Sheet Metal Design | CNC Nesting | Process Improvement | Root Cause Investigation | Production Support | Nonconformance Resolution | Tolerance Stack-Up | Machining (Milling, Drilling, Cutting)

Other: Python | Linux | Trilingual (English, Spanish, Portuguese)